

A PROJECT REPORT ON

“ LeafDoc: AI-Based Plant Identification and Disease Diagnosis System ”

Submitted to

**KDKCE in a partial fulfillment of the requirements for
the award of Master Degree in Computer Application
(M.C.A)**

Submitted by

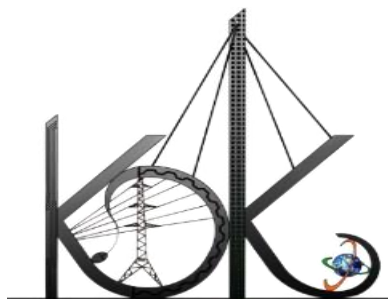
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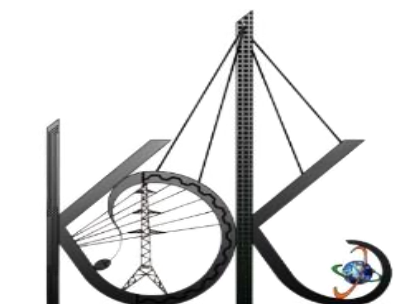
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Department of Computer Application

Session 2025-26



CERTIFICATE

This is to certify that the project work entitled “**LeafDoc: AI-Based Plant Identification and Disease Diagnosis System**” has been carried out by **Ms. Shiwani Amrute and Mr. Prayas Gotefode** of IV Semester M.C.A. and is a bonafide work done under our supervision.

This project work is submitted to KDKCE (An Autonomous Institute), Nagpur in partial fulfillment of the requirement for the award of Master Degree in Computer Application (M.C.A) during the academic year 2025-26.

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Last but not the least we would like to thank entire Department of Computer Application, our parents and all our friends who have helped us in completing this task successfully.

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DECLARATION

The Project work entitled “**LeafDoc: AI-Based Plant Identification and Disease Diagnosis System**” is our own work carried out under the supervision of **Prof. Rahul Lilhare** at KDK College of Engineering, Nagpur. It is ensured that proper citation of references is done.

As far as our knowledge, this work in the same form or any other form is not submitted by us or anyone else for award of any degree.

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ABSTRACT

Accurate plant identification and early disease detection play a vital role in agriculture, horticulture, and home gardening, as plant diseases can significantly reduce crop yield and quality. However, many existing systems require prior knowledge of plant species, which limits their usability for non-expert users. To overcome this limitation, this project presents LeafDoc, an AI-based plant identification and disease diagnosis system.

The proposed system allows users to upload images of plant leaves through a web-based interface. A generative vision-language AI model is used to analyze the image and automatically identify the plant species. Once identified, the system further examines the image to detect possible diseases based on visible symptoms. It then generates meaningful outputs, including plant details, disease information, treatment suggestions, and plant care recommendations.

In addition to analysis, the system integrates an AI-powered chatbot that enables users to interactively ask plant-related queries and receive contextual guidance. The system is designed to be user-friendly, accessible, and efficient, making it suitable for farmers, students, and gardening enthusiasts. By combining plant identification, disease detection, and advisory support into a single platform, LeafDoc provides a practical solution for intelligent plant health monitoring.

KEYWORDS: Plant Identification, Plant Disease Detection, Generative AI, Computer Vision, Smart Agriculture, AI Chatbot, Image Analysis

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CHAPTER - I

INTRODUCTION

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The LeafDoc is a web-based application designed to assist users in plant identification and disease diagnosis using artificial intelligence. In agriculture, horticulture, and home gardening, maintaining plant health is essential for ensuring productivity, sustainability, and environmental balance. Plants are highly susceptible to various diseases caused by bacteria, fungi, viruses, and environmental factors, which can significantly reduce crop yield and quality if not detected at an early stage.

Traditionally, plant identification and disease diagnosis rely on manual observation and expert knowledge. Farmers and gardeners often depend on agricultural experts or personal experience to identify plant species and diagnose diseases. However, this process can be time-consuming, costly, and not always accessible, especially in rural or remote areas where expert consultation may not be readily available. Moreover, incorrect identification or delayed diagnosis can lead to improper treatment, resulting in plant damage and economic loss.

With the advancement of digital technologies, there has been a growing interest in developing automated systems for plant analysis. Image-based analysis using artificial intelligence has emerged as a powerful approach for recognizing plant species and detecting diseases. These systems utilize visual features such as leaf shape, color, texture, and patterns to identify plants and analyze their health condition. However, many existing solutions require prior knowledge of the plant species or depend on large pre-trained datasets, which limits their usability for general users.

The LeafDoc system addresses these limitations by utilizing a generative vision-language AI model capable of interpreting plant images and generating meaningful outputs. The system allows users to upload images of plant leaves through a simple web interface. Based on the uploaded image, the system first identifies the plant species and then analyzes the image to detect possible diseases. It provides detailed information, including plant name, disease status, treatment suggestions, and plant care guidelines.

One of the key features of LeafDoc is the integration of an AI-powered chatbot that enables interactive communication between the user and the system. Users can ask questions related to plant care, disease prevention, and treatment methods, and the chatbot provides contextual and informative responses. This feature enhances user engagement and makes the system more accessible, especially for non-expert users.

The system is designed to be user-friendly, scalable, and efficient. It eliminates the need for manual feature extraction or complex model training by leveraging API-based generative AI. This approach simplifies the implementation while maintaining high flexibility and adaptability. The web-based nature of the system ensures that users can access it from different devices without requiring installation.

Overall, LeafDoc provides an integrated solution for plant identification, disease diagnosis, and advisory support. It contributes to the field of smart agriculture by offering an intelligent and practical tool that helps users monitor plant health and make informed decisions.

1.2 OBJECTIVE

The objective of the proposed LeafDoc system is to develop an intelligent, efficient, and user-friendly solution for plant identification and disease diagnosis using artificial intelligence. The system aims to act as a support tool that assists users in analyzing plant health through image-based input, thereby reducing dependency on expert knowledge and improving accessibility.

The system is designed to integrate modern technologies such as computer vision and generative artificial intelligence to provide accurate and meaningful results. It focuses on delivering a practical solution that can be used in real-world agricultural and gardening environments.

The specific objectives of the system are as follows:

1. To develop a web-based application for plant identification using image input
2. To automatically identify plant species without requiring prior user knowledge
3. To detect plant diseases based on leaf images and visual symptoms
4. To provide treatment suggestions and plant care recommendations
5. To integrate an AI-powered chatbot for interactive user assistance
6. To design a simple, intuitive, and user-friendly interface

7. To ensure accessibility across multiple devices through a web-based system
8. To develop a scalable system that supports future enhancements

1.3 PROBLEM DEFINATION

Plant health management is a critical aspect of agriculture and gardening, yet it remains a challenging task for many users. One of the major problems faced by farmers, students, and plant enthusiasts is the difficulty in accurately identifying plant species. Many plants have similar visual characteristics, especially during early growth stages, making it difficult to distinguish between them without expert knowledge.

Another significant problem is the detection of plant diseases. Diseases often appear as small changes in leaf color, texture, or structure, which may not be easily noticeable to non-expert users. As a result, diseases are often detected at later stages, when the damage has already spread. This leads to reduced crop yield, poor plant quality, and economic losses.

Existing plant disease detection systems have several limitations. Many systems require users to manually select or input the plant species before performing disease analysis. This creates an additional burden for users who are not familiar with plant taxonomy. Furthermore, most systems focus only on disease detection and do not provide complete solutions such as treatment recommendations or care guidelines.

Traditional methods of plant diagnosis involve manual inspection and consultation with agricultural experts. While these methods can be accurate, they are not always practical due to time constraints, cost, and limited availability of experts. In rural areas, access to such expertise is often restricted, making it difficult for farmers to obtain timely assistance.

Another challenge is the lack of integrated systems that combine plant identification, disease detection, and advisory support. Users often need to rely on multiple sources to gather information, which can be inefficient and confusing. Additionally, the absence of interactive guidance limits the usability of existing systems.

Therefore, there is a need for a unified system that can automatically identify plants, detect diseases, and provide actionable recommendations in a simple and accessible manner. The proposed LeafDoc system aims to address these challenges by offering an integrated and AI-driven solution for plant health monitoring.

CHAPTER - II

RELATED WORK

CHAPTER 2

RELATED WORK

2.1 LITERATURE REVIEW

In recent years, advancements in digital technologies and artificial intelligence have significantly transformed the field of agriculture and plant health monitoring. The increasing demand for efficient and automated solutions has led to the development of various systems for plant identification and disease detection. Researchers have explored different approaches, including image processing, machine learning, and deep learning techniques, to analyze plant images and identify diseases.

One of the most widely studied approaches is the use of Convolutional Neural Networks (CNNs) for plant disease detection. Studies by S. P. Mohanty et al. [11] demonstrated that deep learning models can achieve high accuracy in classifying plant diseases from leaf images. These models are trained on large datasets containing labeled images of healthy and diseased plants, enabling them to learn complex visual patterns. However, such approaches require extensive datasets and significant computational resources, which can limit their practical deployment.

Similarly, K. P. Ferentinos [10] conducted a comparative study of various deep learning architectures and showed that deep learning models outperform traditional machine learning techniques in plant disease classification. Despite their effectiveness, these systems often depend on prior knowledge of plant species, requiring users to specify the plant type before disease detection.

In the area of plant identification, image-based techniques have been widely used to recognize plant species using features such as leaf shape, texture, and color. Deep learning-based approaches, such as those discussed by M. Sladojevic et al. [12], have improved the accuracy of plant disease recognition. However, these systems are generally limited to classification tasks and do not provide integrated advisory support.

In addition to detection systems, decision-support systems have been developed to assist users in agricultural decision-making. R. Ramesh and R. S. K. M. Reddy [14] proposed systems that provide recommendations based on predefined rules and expert knowledge. While useful, these systems lack image-based automation and require manual input from users.

Recent advancements in artificial intelligence, particularly generative AI and vision-language models, have introduced new possibilities for image interpretation. Research by A. Radford et al. [5] and T. Brown et al. [7] demonstrated that such models can interpret visual data and generate meaningful textual outputs without requiring task-specific training. These models enable flexible and scalable solutions for integrated systems.

Despite these advancements, most existing systems focus on individual tasks such as plant identification or disease detection. There is limited research on integrating these functionalities into a single system that also provides advisory support. The proposed LeafDoc system aims to address this gap by combining plant identification, disease diagnosis, and recommendation generation into a unified platform using generative AI.

2.2 BACKGROUND STUDY

The development of the LeafDoc system is based on fundamental concepts in artificial intelligence, computer vision, and modern agricultural technologies. Understanding these concepts is essential for designing an efficient, scalable, and reliable system for plant identification and disease diagnosis.

Artificial Intelligence in Agriculture

Artificial intelligence has become a significant component in modern agriculture, enabling automation and intelligent decision-making. It is widely used in applications such as crop monitoring, yield prediction, pest detection, and disease diagnosis. AI-based systems analyze large volumes of agricultural data to generate accurate and timely insights, which help improve productivity and resource management [10], [14].

Image Processing and Computer Vision

Image processing plays a crucial role in plant analysis by enhancing and analyzing images to extract meaningful information. Computer vision extends these capabilities by allowing machines to interpret and understand visual data. In plant disease detection, computer vision techniques are used to analyze leaf images and identify patterns such as discoloration, spots, and texture variations that indicate disease presence [11], [12].

Deep Learning Techniques

Deep learning, a subset of machine learning, utilizes neural networks with multiple layers to learn complex patterns from data. Convolutional Neural Networks (CNNs) are widely used for image classification tasks, including plant disease detection. These models automatically extract relevant features from images and classify them with high accuracy. The use of deep learning has significantly enhanced the performance of image-based plant analysis systems [11], [6].

Generative AI and Vision-Language Models

Generative artificial intelligence represents a recent advancement in AI, where models can generate meaningful outputs such as text, descriptions, and recommendations. Vision-language models combine computer vision with natural language processing, enabling them to analyze images and produce descriptive and contextual outputs. Unlike traditional models, these systems do not require extensive task-specific training, making them flexible and suitable for integrated applications like LeafDoc [5], [7].

Web-Based Systems

Web-based applications provide a convenient and accessible platform for users to interact with the system. They allow users to upload images, view analysis results, and access recommendations from any device with an internet connection. This approach ensures scalability, ease of deployment, and improved user accessibility.

Role of Advisory Systems

Advisory systems play an important role in supporting decision-making processes in agriculture. These systems provide recommendations for plant care, disease treatment, and preventive measures. Integrating advisory capabilities into plant analysis systems enhances their usability and makes them more practical for real-world applications [14].

In conclusion, the LeafDoc system integrates these fundamental concepts to develop a comprehensive and intelligent solution for plant identification and disease diagnosis. By combining artificial intelligence, computer vision, and web technologies, the system provides an efficient and user-friendly platform for plant health monitoring.

2.3 RELATED WORK

Several research works and systems have been developed for plant identification and disease detection using artificial intelligence and image processing techniques. These systems have contributed significantly to agricultural automation; however, they still exhibit limitations in terms of usability, integration, and practical applicability.

Deep learning-based approaches have been widely used for plant disease detection. S. P. Mohanty et al. [11] demonstrated that convolutional neural networks can achieve high accuracy in classifying plant diseases from leaf images. Similarly, K. P. Ferentinos [10] evaluated various deep learning architectures and reported improved performance compared to traditional methods. Although these systems are highly accurate, they require large labeled datasets and assume that the plant species is already known, which limits their usability for non-expert users.

In the area of plant identification, image-based systems have been developed to recognize plant species using visual features such as leaf shape, color, and texture. M. Sladojevic et al. [12] proposed a deep learning-based approach for plant disease recognition using leaf images. While such systems provide reliable classification results, they are generally limited to specific tasks and do not integrate plant identification, disease detection, and advisory support within a single framework.

Mobile-based plant analysis systems have also gained popularity due to their accessibility and ease of use. P. Picon et al. [8] introduced a smartphone-based system for crop disease classification using image input. Although these systems improve usability and enable real-time analysis, they often require manual input of plant information and lack detailed treatment recommendations.

Agricultural decision-support systems have been developed to assist users in plant care and disease management. R. Ramesh and R. S. K. M. Reddy [14] proposed systems that provide recommendations based on expert knowledge and predefined rules. However, these systems do not incorporate image-based analysis and rely heavily on user input, which may reduce accuracy in real-world scenarios.

Recent advancements in artificial intelligence, particularly vision-language models, have introduced new possibilities for image interpretation. A. Radford et al. [5] and T. Brown et al.

[7] demonstrated that such models can analyze visual data and generate meaningful textual outputs without task-specific training. While these models offer flexibility and scalability, their application in integrated plant health monitoring systems is still limited.

From the above studies, it is evident that most existing systems focus on individual functionalities such as plant identification or disease detection. Additionally, they lack integrated advisory support and interactive user interfaces. Therefore, there is a need for a unified system that combines plant identification, disease diagnosis, and recommendation generation within a single platform. The proposed LeafDoc system addresses these limitations by providing an intelligent, user-friendly, and integrated solution for plant health monitoring.

Sr. No.	Author (Ref.)	Paper Title	Year	Method / Approach	Key Contribution	Limitation
1	A. Radford et al. [5]	Learning transferable visual models	2021	Vision-Language Model	Learns visual-text relationships	Limited agriculture use
2	T. Brown et al. [7]	Language models are few-shot learners	2020	Generative AI Model	Enables text-based AI interaction	Not domain-specific
3	P. Picon et al. [8]	Crop disease classification using smartphone images	2019	Mobile-based System	Real-time plant disease detection	Requires manual input
4	K. P. Ferentinos [10]	Deep learning models for plant disease detection	2018	Deep Learning Models	Improved classification performance	Lacks user guidance

Sr. No.	Author (Ref.)	Paper Title	Year	Method / Approach	Key Contribution	Limitation
5	S. P. Mohanty et al. [11]	Using deep learning for image-based plant disease detection	2016	CNN-based Deep Learning	High accuracy in plant disease detection	Requires large datasets and prior knowledge
6	M. Sladojevic et al. [12]	Deep neural networks for plant disease recognition	2016	Deep Neural Networks	Effective disease classification	No advisory support
7	R. Ramesh and R. S. K. M. Reddy [14]	Decision support systems in agriculture	2014	Rule-based System	Provides expert recommendations	No image-based analysis

Table 2.1: Related Work

CHAPTER - III

PROPOSED SYSTEM

CHAPTER 3

PROPOSED SYSTEM

3.1 EXISTING SYSTEM

In the current scenario, plant identification and disease detection are primarily carried out using traditional methods or partially automated systems. Farmers, gardeners, and plant enthusiasts often rely on manual observation and personal experience to identify plant species and diagnose diseases. This process requires domain knowledge and expertise, which may not be available to all users.

One common method used in practice is visual inspection. Users examine plant leaves for visible symptoms such as discoloration, spots, or deformation. Based on these observations, they attempt to identify the disease. However, this approach is highly subjective and prone to errors, especially for non-expert users. Many plant diseases exhibit similar symptoms, making it difficult to accurately diagnose them without proper knowledge.

Another approach involves consulting agricultural experts or referring to books and online resources. While this method can provide accurate results, it is time-consuming and not always accessible. In rural areas, farmers may not have easy access to experts, and delays in diagnosis can lead to the spread of diseases and reduced crop yield.

In recent years, several digital solutions have been developed for plant identification and disease detection. Many of these systems are based on machine learning or deep learning models trained on plant image datasets. These systems can analyze images and classify diseases with high accuracy. However, most of these systems require users to input or select the plant species before performing disease detection.

Additionally, some mobile applications provide plant identification features using image recognition. These applications allow users to capture images and receive plant information. However, they often focus only on identification and do not provide detailed disease analysis or treatment recommendations.

Agricultural advisory systems also exist, which provide guidance based on environmental conditions and predefined rules. These systems help users make decisions related to irrigation,

fertilization, and pest control. However, they lack image-based analysis and require manual input from users, which may not always be accurate.

Overall, the existing systems are fragmented and lack integration. Users often need to rely on multiple tools or platforms to perform plant identification, disease diagnosis, and obtain recommendations. This creates inefficiency and reduces usability, especially for non-expert users.

3.2 LIMITATIONS OF EXISTING SYSTEM

The existing systems for plant identification and disease detection have several limitations that affect their usability and effectiveness. These limitations highlight the need for a more advanced and integrated solution.

1. **Lack of Integrated Functionality:**

Most existing systems focus on a single task, such as plant identification or disease detection. Users must use separate tools for different purposes, which makes the process time-consuming and inefficient.

2. **Dependency on Prior Knowledge:**

Many systems require users to know the plant species before performing disease detection. This creates a challenge for non-expert users who may not be familiar with plant taxonomy.

3. **Limited Accessibility:**

Traditional methods such as expert consultation are not easily accessible, especially in rural or remote areas. This limits the ability of users to obtain timely assistance.

4. **Time-Consuming Process:**

Manual diagnosis and consultation require significant time and effort. Delays in identifying plant diseases can result in increased damage and reduced productivity.

5. **Lack of Real-Time Analysis:**

Most systems do not provide instant results. Users may need to wait for expert responses or perform multiple steps to obtain information.

6. **No Advisory Support:**

Many systems only provide identification or classification results without offering treatment recommendations or plant care guidance.

7. Data Dependency and Training Requirements:

Deep learning models require large datasets and training, which increases complexity and limits scalability.

8. User Interface Complexity:

Some existing applications have complex interfaces that are not user-friendly, making them difficult to use for beginners.

9. Inaccuracy Due to Manual Input:

Systems that rely on user input for symptoms or plant details may produce incorrect results if the input is inaccurate.

These limitations indicate the need for a system that is simple, integrated, and capable of providing accurate results without requiring extensive user input or expertise.

3.3 PROPOSED SYSTEM

The proposed system, LeafDoc, is a Generative AI-based plant identification and disease diagnosis system designed to overcome the limitations of existing approaches. It provides a unified platform that integrates plant identification, disease detection, and advisory support into a single system.

The system allows users to upload images of plant leaves through a web-based interface. The uploaded image is analyzed using a generative vision-language AI model, which interprets the visual data and generates structured outputs. The system first identifies the plant species and then analyzes the image for possible diseases based on visible symptoms.

Unlike traditional systems, LeafDoc does not require prior knowledge of plant species. This makes it highly accessible to non-expert users. The system automatically performs identification and diagnosis in a sequential manner, ensuring accurate and relevant analysis.

Once the disease is detected, the system provides treatment suggestions and plant care recommendations. These recommendations help users take appropriate actions to improve plant health and prevent further damage.

In addition to analysis, the system includes an AI-powered chatbot that allows users to ask questions related to plant care and disease prevention. The chatbot provides contextual responses, enhancing user interaction and support.

The system is implemented as a web-based application, ensuring easy accessibility across different devices. It uses secure API-based integration for AI processing, eliminating the need for complex model training and dataset management.

Overall, the proposed system offers a practical, efficient, and user-friendly solution for plant health monitoring, making it suitable for farmers, students, and gardening enthusiasts.

3.4 FEATURES OF PROPOSED SYSTEM

The LeafDoc system includes several advanced features that enhance its functionality and usability:

1. **Automated Plant Identification:**

The system automatically identifies plant species from uploaded images without requiring user input.

2. **Disease Detection:**

The system analyzes leaf images to detect possible diseases based on visual symptoms.

3. **AI-Based Recommendations:**

Provides treatment suggestions and plant care guidelines to help users manage plant health.

4. **Integrated Chatbot:**

An AI-powered chatbot provides interactive assistance and answers plant-related queries.

5. **User-Friendly Interface:**

The system features a simple and intuitive interface for easy navigation and usage.

6. **Web-Based Accessibility:**

Users can access the system from any device with an internet connection.

7. **No Training Required:**

Uses generative AI, eliminating the need for dataset collection and model training.

8. **Scalable Design:**

The system can be extended with additional features in the future.

9. **Real-Time Analysis:**

Provides instant results for plant identification and disease detection.

3.5 FEASIBILITY STUDY

The feasibility study evaluates whether the proposed system is practical and viable for implementation.

3.5.1 Technical Feasibility

The LeafDoc system is technically feasible as it uses modern web technologies and AI integration. The system does not require complex hardware or high computational resources, as the AI processing is handled through API-based services. The use of web technologies ensures compatibility across different platforms and devices.

3.5.2 Operational Feasibility

The system is designed to be user-friendly and easy to operate. Users do not require technical knowledge to use the system. The simple interface and automated analysis make it suitable for a wide range of users, including farmers, students, and beginners.

3.5.3 Economic Feasibility

The system is economically viable as it uses open-source technologies and API-based services. The cost of development and maintenance is relatively low compared to traditional systems. This makes it suitable for deployment in educational and agricultural environments.

CHAPTER - IV

METHODOLOGY

CHAPTER 4

METHODOLOGY

4.1 METHODOLOGY

The methodology adopted for the development of the LeafDoc system focuses on creating an efficient, scalable, and user-friendly solution for plant identification and disease diagnosis using artificial intelligence. The system is designed using a structured and modular approach to ensure proper organization, maintainability, and smooth workflow execution.

Unlike traditional machine learning systems that require extensive dataset collection, preprocessing, and model training, the LeafDoc system utilizes a generative vision-language AI model through API-based integration. This approach eliminates the need for manual feature extraction and reduces system complexity while maintaining flexibility and adaptability.

The development methodology follows a sequence of stages including requirement analysis, system design, implementation, and testing. Initially, the requirements of the system are identified, focusing on user needs such as plant identification, disease detection, and advisory support. Based on these requirements, the system architecture and workflow are designed.

The system follows a client–server architecture where the frontend handles user interaction and the backend manages communication with the AI model. The user uploads an image through the web interface, which is then processed by the backend and sent to the AI model for analysis. The AI model interprets the image and generates structured outputs such as plant identification, disease diagnosis, and recommendations.

This methodology ensures that the system remains simple for users while maintaining powerful analytical capabilities. The modular design also allows easy integration of additional features such as mobile applications, multilingual support, and real-time monitoring in future enhancements.

The system incorporates data validation and preprocessing mechanisms to ensure reliable analysis. When a user uploads an image, the system verifies the format, size, and clarity of the image before forwarding it for AI processing. This step helps in minimizing errors and improving the accuracy of the results. The backend also manages request handling efficiently

by organizing inputs and responses in a structured manner. This ensures that the interaction between the user interface and the AI model remains smooth and consistent.

The system emphasizes usability and performance optimization. The user interface is designed to provide quick responses and clear outputs, reducing the complexity for non-technical users. The integration of the chatbot enhances the methodology by enabling continuous interaction and guidance beyond image-based analysis. The system is also designed to handle multiple requests efficiently, ensuring scalability for future deployment. This comprehensive methodology ensures that LeafDoc not only performs accurate plant analysis but also delivers a seamless and user-friendly experience.

4.2 SYSTEM WORKFLOW

The workflow of the LeafDoc system is designed to ensure smooth and efficient processing of user inputs and generation of outputs. The workflow follows a sequential process where each stage contributes to the overall functionality of the system.

Workflow Steps:

1. **Image Upload:**

The user uploads an image of a plant or leaf through the web interface. The system accepts standard image formats and performs basic validation.

2. **Image Processing:**

The uploaded image is prepared and formatted for analysis. The system ensures that the image is suitable for AI processing.

3. **Plant Identification:**

The image is sent to the generative AI model, which analyzes visual features and identifies the plant species.

4. **Disease Detection:**

After identifying the plant, the system analyzes the image for possible diseases based on visible symptoms.

5. **Recommendation Generation:**

The system generates treatment suggestions and plant care guidelines based on the identified disease.

6. Result Display:

The results are displayed to the user in a structured and easy-to-understand format.

7. Chatbot Interaction:

The user can interact with the chatbot to ask additional questions related to plant care.

This workflow ensures a seamless and efficient user experience.

4.3 SYSTEM ARCHITECTURE

A The LeafDoc system follows a three-tier architecture, which separates the system into different layers for better organization and performance.

1. Presentation Layer (Frontend)

This layer handles user interaction. It includes web pages such as the home page, identification page, and chatbot interface. The frontend is responsible for collecting user input and displaying results.

2. Application Layer (Backend)

This layer contains the core logic of the system. It processes user requests, communicates with the AI model, and generates responses. It acts as a bridge between the frontend and the AI services.

3. AI Processing Layer

This layer consists of the generative vision-language AI model. It analyzes the uploaded images and generates outputs such as plant identification, disease diagnosis, and recommendations.

Layer	Component	Function
Presentation	User Interface	Allows image upload and displays results
Application	Backend Server	Handles requests and communicates with AI
AI Processing	AI Model	Performs plant identification and disease detection

Table 4.1: System Architecture Layers

LeafDoc System Architecture

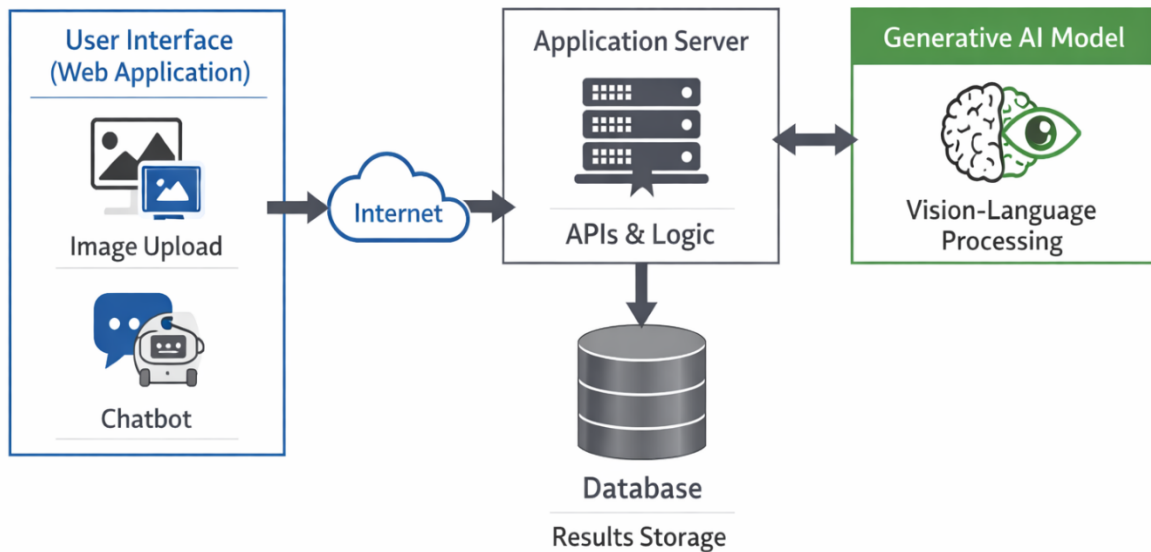


Figure 4.1: System Architecture

4.4. HARDWARE COMPONENTS

The LeafDoc system requires minimal hardware resources as it is a web-based application.

Server-Side Requirements

- Processor: Dual-core or higher
- RAM: Minimum 4 GB
- Storage: 50 GB
- Internet: Required

Client-Side Requirements

- Device: Smartphone / Laptop / PC
- Browser: Chrome, Edge, Firefox
- Internet: Required

Development Requirements

- System: Laptop/PC
- Tools: Visual Studio Code
- Technologies: HTML, CSS, JavaScript, Backend framework

Category	Component	Requirement
Server	Processor	Dual-core or higher
Server	RAM	Minimum 4 GB
Server	Storage	50 GB
Client	Device	Smartphone / PC
Client	Browser	Modern browser
Development	Tools	VS Code

Table 4.2: Hardware Requirements

Hardware Components

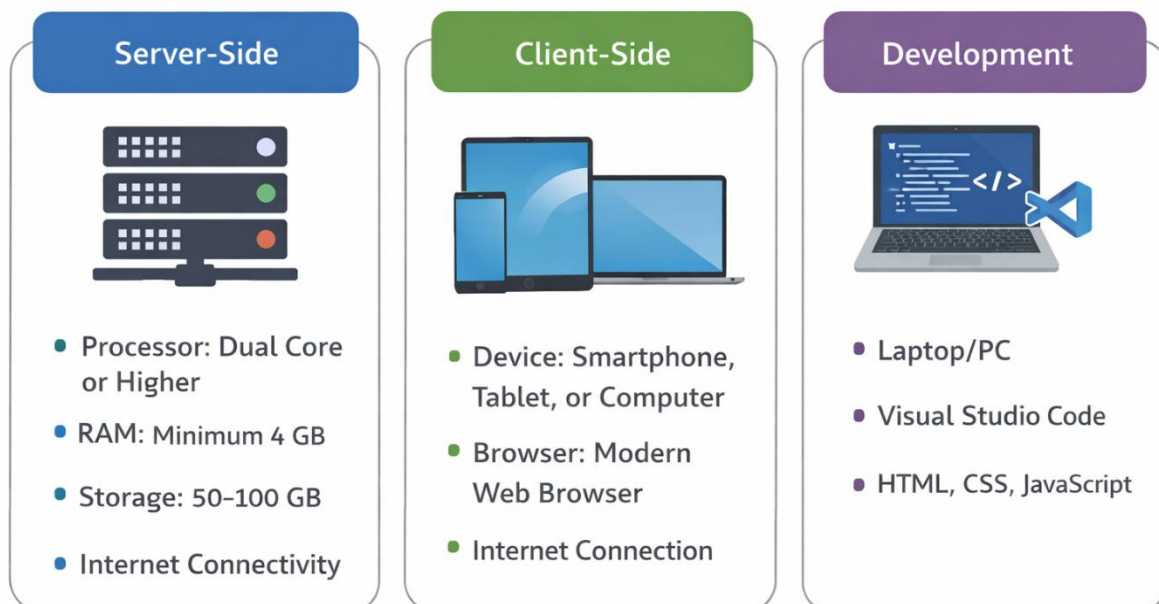


Figure 4.2: Hardware Component

4.5 PROCESS FLOW (ALGORITHM)

The process flow of the system describes the step-by-step execution of the LeafDoc system.

Algorithm Steps:

1. Start
2. User uploads plant image
3. System validates the image
4. Image is sent to AI model
5. AI model analyzes the image
6. Identify plant species
7. Detect possible disease
8. Generate recommendations
9. Display results to user
10. Provide chatbot support
11. End

This algorithm ensures a structured and efficient execution of the system.

The above process flow represents a structured and logical sequence of operations carried out within the LeafDoc system. Each stage of the algorithm contributes to accurate plant identification and disease detection by ensuring proper handling of user input, efficient communication with the AI model, and systematic generation of outputs. The validation step improves reliability, while the integration of AI processing enables intelligent analysis of plant images. Furthermore, the inclusion of chatbot interaction enhances user support by providing additional guidance beyond the initial results. Overall, the process flow ensures efficient system performance, accuracy, and a user-friendly experience.

LeafDoc System Flowchart

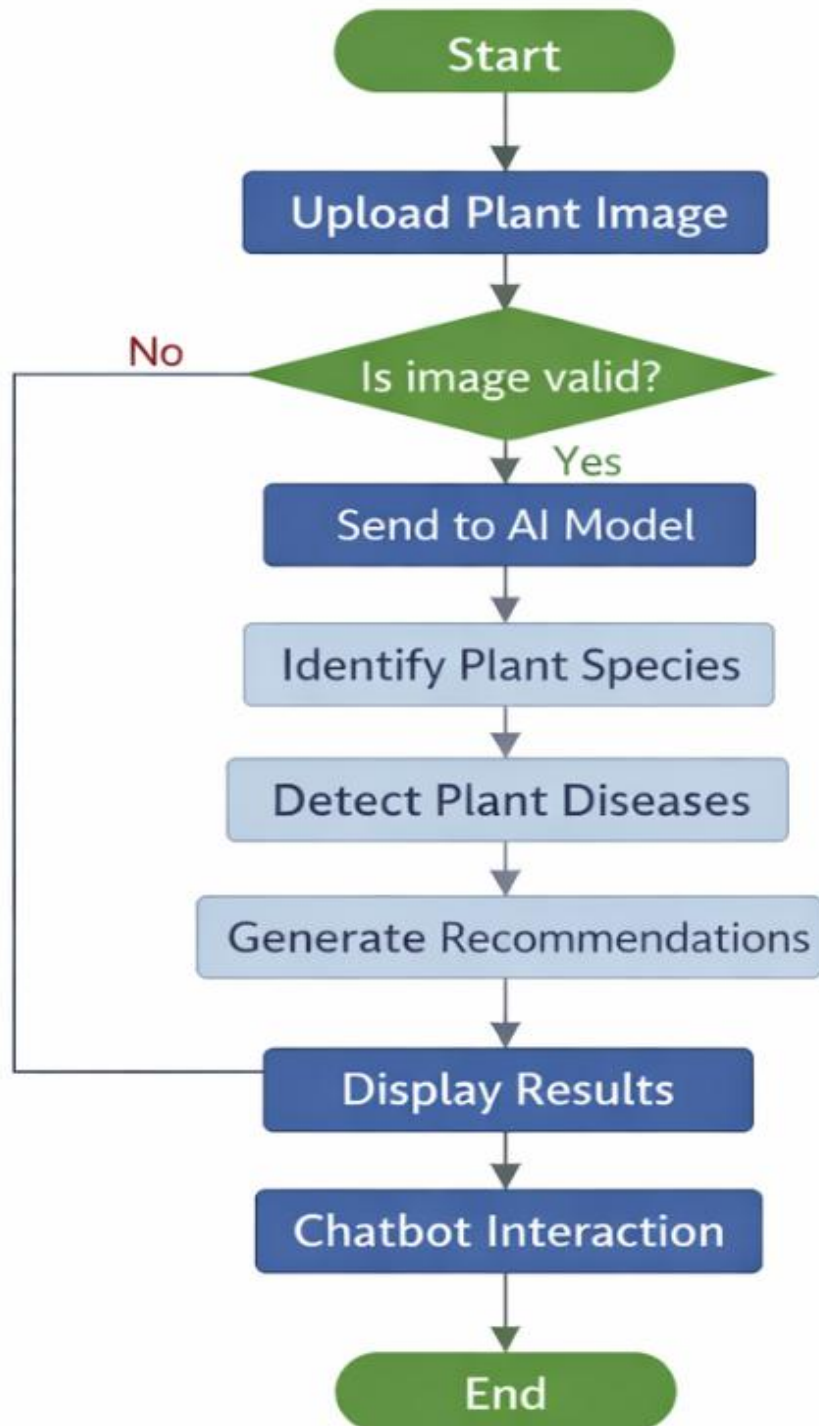


Figure 4.3: Process Flow

CHAPTER - V
EXPERIMENTAL SETUP &
RESULTS

CHAPTER 5

EXPERIMENTAL SETUP & RESULTS

5.1 EXPERIMENTAL RESULTS

The LeafDoc system was tested under various conditions to evaluate its performance in plant identification and disease detection. Different plant images, including both healthy and diseased samples, were used as input to assess the accuracy and effectiveness of the system.

The system successfully identified plant species from uploaded images and provided relevant disease detection results along with treatment recommendations. In most cases, the outputs generated by the AI model were accurate and meaningful, demonstrating the capability of the system to analyze plant images effectively.

The response time of the system was also observed to be efficient, as the API-based AI model processed the images quickly and returned results within a short duration. The system performed well across different devices such as laptops and smartphones, confirming its accessibility and usability.

Additionally, the chatbot feature was tested by providing various plant-related queries. The chatbot was able to generate appropriate and informative responses, enhancing the overall user experience.

However, it was observed that the accuracy of the system depends on the quality of the input image. Clear and high-resolution images resulted in better outputs, while low-quality images sometimes affected the accuracy of detection.

Overall, the experimental results indicate that the LeafDoc system performs effectively in real-world scenarios and meets the intended objectives of plant identification, disease detection, and advisory support.

5.2 IMPLEMENTATION TOOL

The development of the LeafDoc system involves a combination of frontend, backend, and artificial intelligence technologies that work together to provide a complete and efficient solution.

Frontend Technologies

HTML, CSS, and JavaScript are used to design and develop the user interface. These technologies enable the creation of responsive and interactive web pages, allowing users to easily upload images and view results.

Backend Technologies

Node.js is used for handling server-side logic and managing communication between the user interface and the AI model. The backend processes user requests and ensures smooth data flow within the system.

Artificial Intelligence

A generative vision-language AI model is integrated into the system for plant identification and disease detection. It analyzes uploaded images and generates structured outputs such as plant details, disease information, and recommendations.

Development Tools

Visual Studio Code is used for coding, debugging, and testing the application. It provides a flexible environment for efficient development and project management.

The system utilizes secure API-based communication to connect the frontend and backend with the AI model. Environment variables are used to manage sensitive data such as API keys, ensuring system security. The selected technologies provide flexibility and scalability, allowing future enhancements such as mobile integration and additional features. Overall, the implementation ensures a robust, efficient, and user-friendly application for plant health monitoring.

5.3 SCREENSHOTS OF RESULTS

The following screenshots represent the output screens captured during system execution. These images demonstrate the working modules and validate the successful implementation of the system

1. Home Page:

The home page provides an overview of the LeafDoc system, highlighting its purpose of plant identification and disease diagnosis using AI. It offers a clean and user-friendly interface with navigation options for accessing different features of the application.

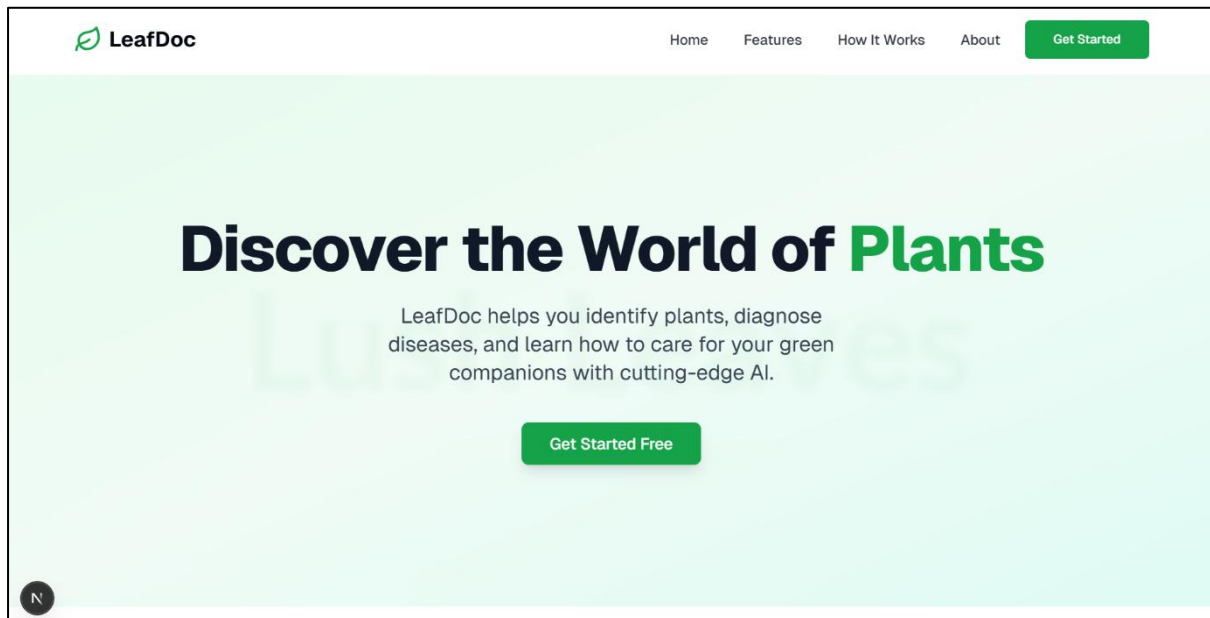


Figure 5.1: Home Page

2. Features Page:

The features page showcases the core functionalities of the system, including plant identification, disease detection, and care guidance. It helps users understand the capabilities of LeafDoc in a simple and visually appealing manner.

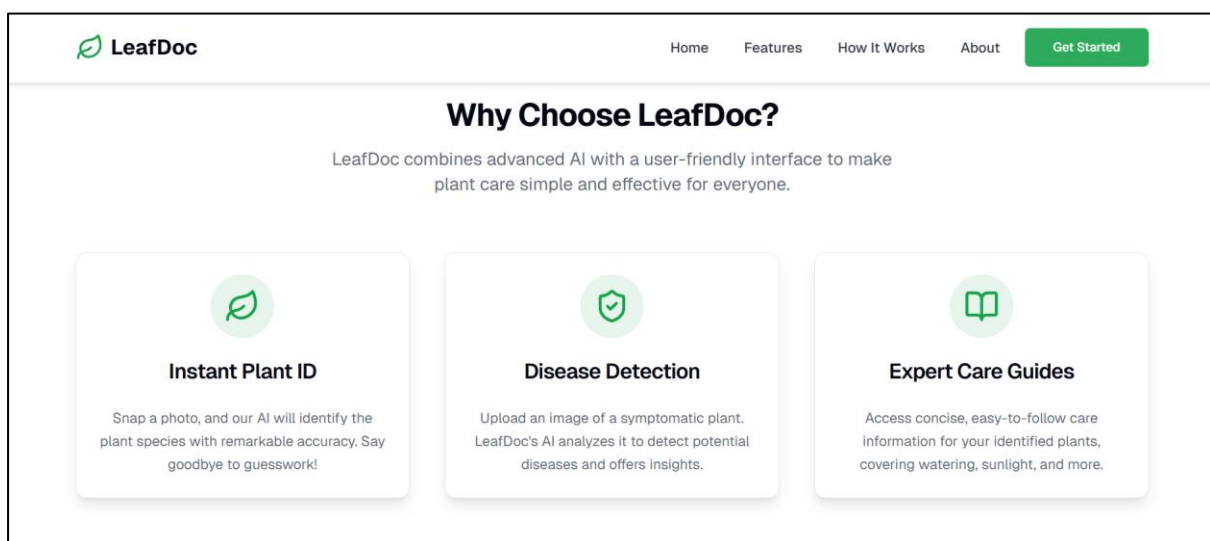


Figure 5.2: Features

3. Plant Identification Module:

This module allows users to upload or capture plant images for identification. The AI model analyzes visual features and accurately determines the plant species along with relevant details.

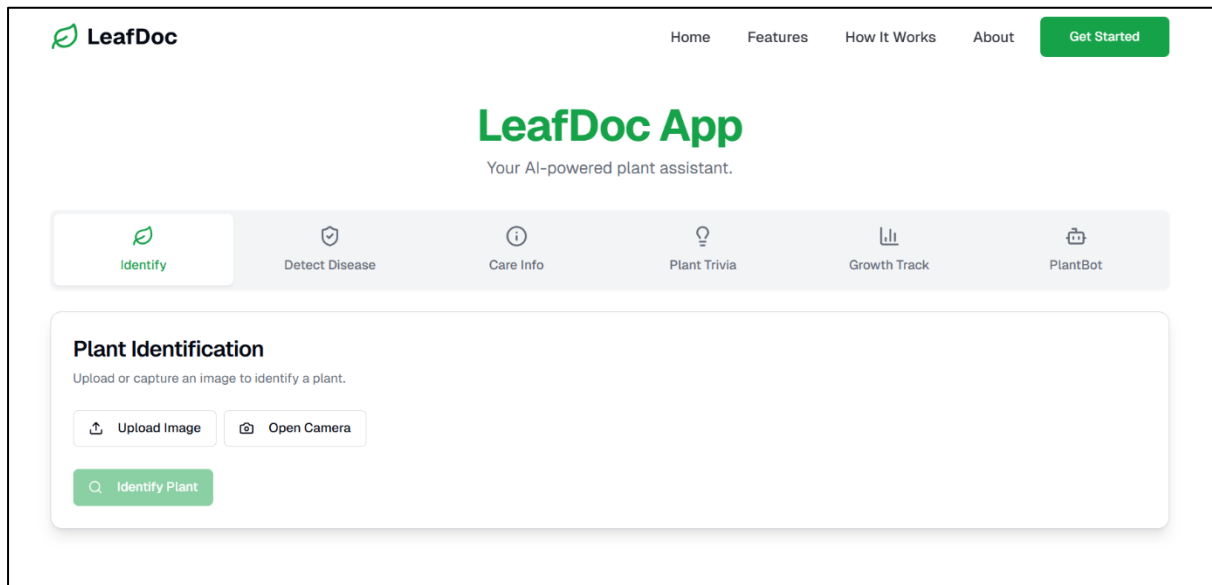


Figure 5.3: Identification

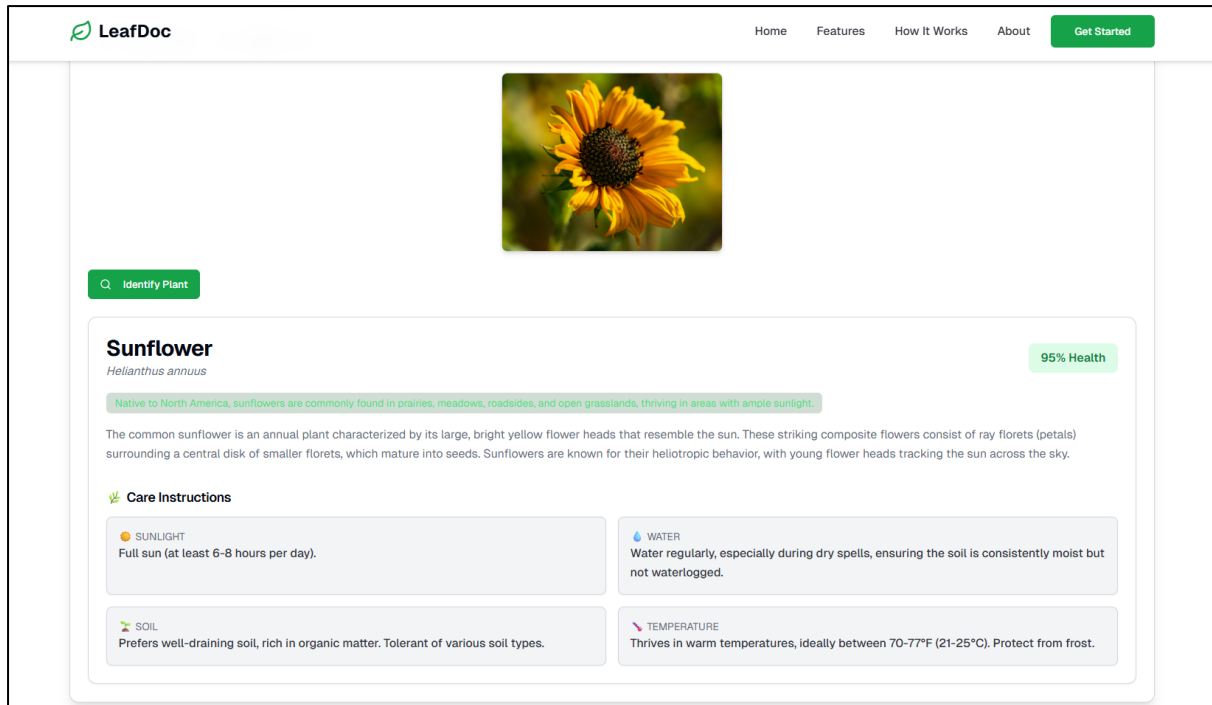
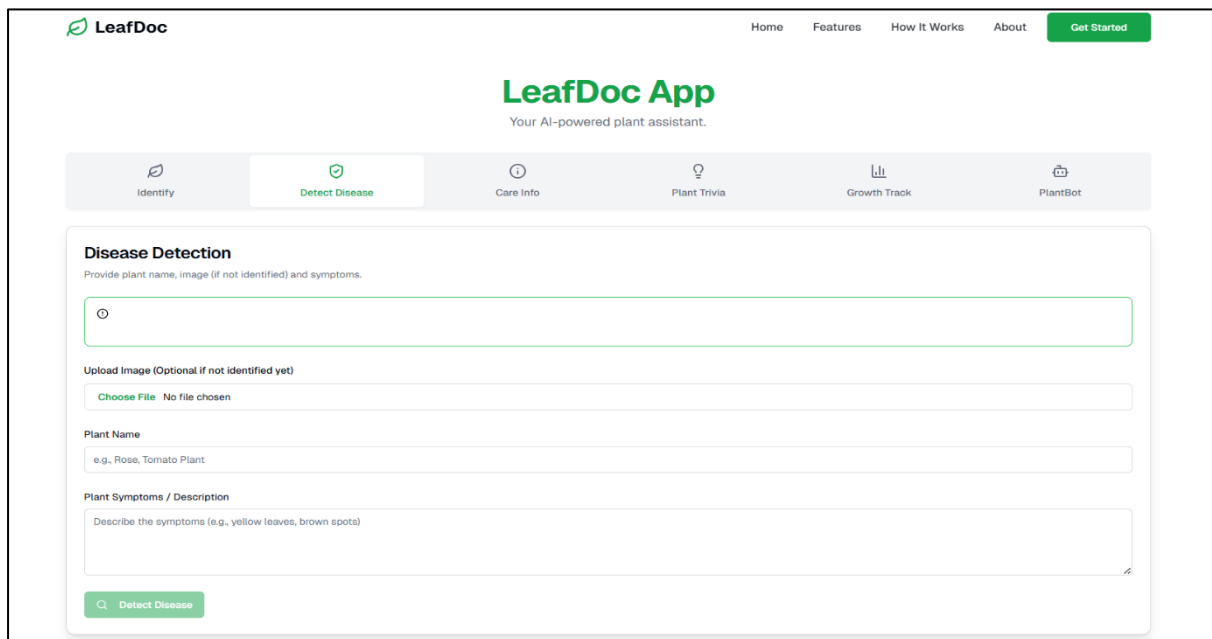


Figure 5.4: Identification Output

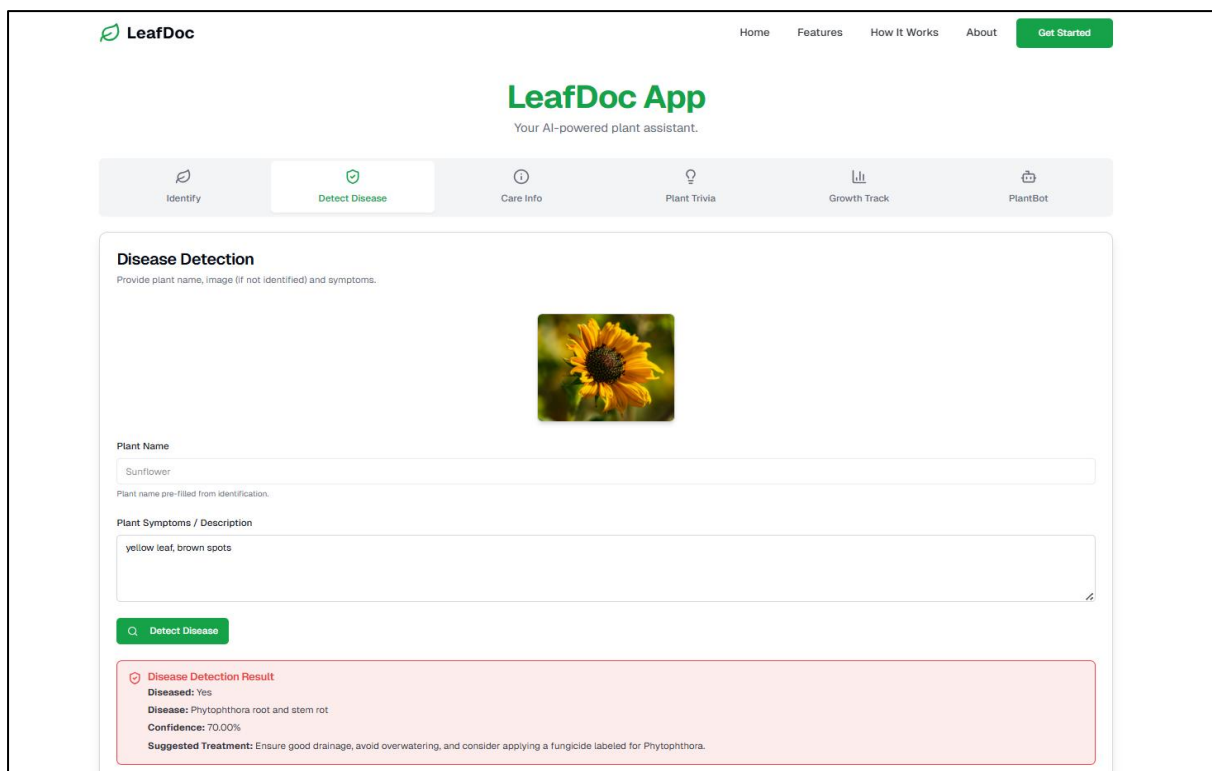
4. Disease Detection Module:

The disease detection module analyzes plant symptoms and images to identify possible diseases. It provides disease name, confidence level, and treatment suggestions for effective plant care.



The screenshot shows the LeafDoc App interface. At the top, there's a navigation bar with links: Home, Features, How It Works, About, and a green 'Get Started' button. Below the navigation bar, the app's name 'LeafDoc App' is displayed with the tagline 'Your AI-powered plant assistant.' A horizontal menu contains six icons: Identify, Detect Disease (highlighted in green), Care Info, Plant Trivia, Growth Track, and PlantBot. The main content area is titled 'Disease Detection' and includes the instruction 'Provide plant name, image (if not identified) and symptoms.' There are three input fields: a large text area for the plant name (containing a placeholder icon), an 'Upload Image' section with a 'Choose File' button and 'No file chosen' text, and a 'Plant Name' field with the example text 'e.g., Rose, Tomato Plant'. Below these is a 'Plant Symptoms / Description' text area with the placeholder 'Describe the symptoms (e.g., yellow leaves, brown spots)'. At the bottom of the form is a green 'Detect Disease' button.

Figure 5.5: Disease Detection



This screenshot shows the same LeafDoc App interface as Figure 5.5, but with the 'Detect Disease' button clicked. The 'Plant Name' field now contains 'Sunflower' and is followed by the text 'Plant name pre-filled from identification.' The 'Plant Symptoms / Description' field contains 'yellow leaf, brown spots'. Below the input fields is a green 'Detect Disease' button. At the bottom of the form, a red-bordered box displays the 'Disease Detection Result'. The results are as follows:

Disease Detection Result	
Diseased:	Yes
Disease:	Phytophthora root and stem rot
Confidence:	70.00%
Suggested Treatment:	Ensure good drainage, avoid overwatering, and consider applying a fungicide labeled for Phytophthora.

Figure 5.6: Disease Detection Output

5. Care Information Module:

This module provides detailed plant care guidelines such as watering, sunlight, and maintenance tips. It helps users maintain plant health based on identified species.

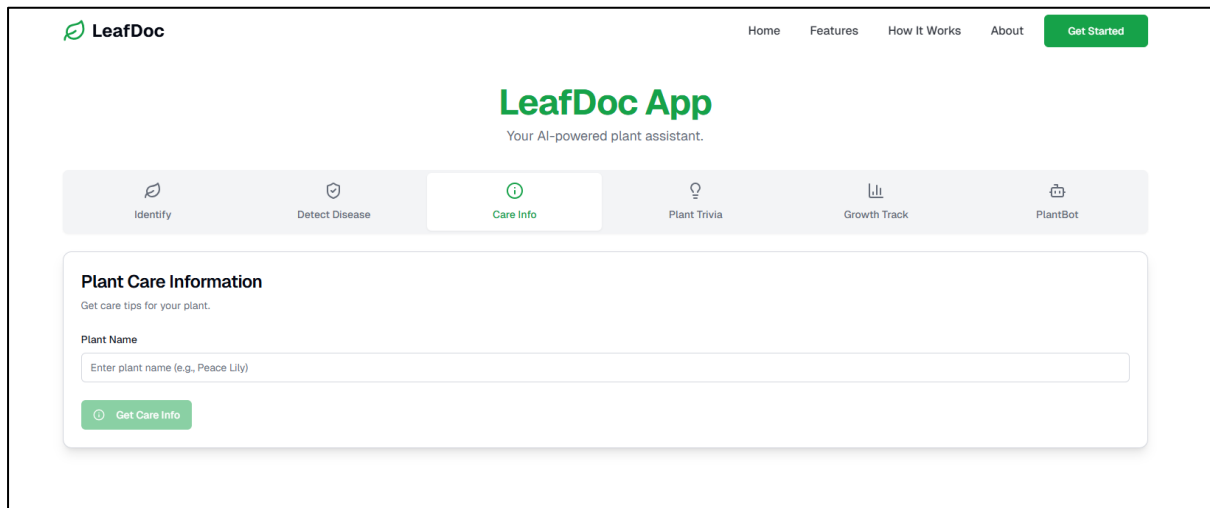


Figure 5.7: Care Information

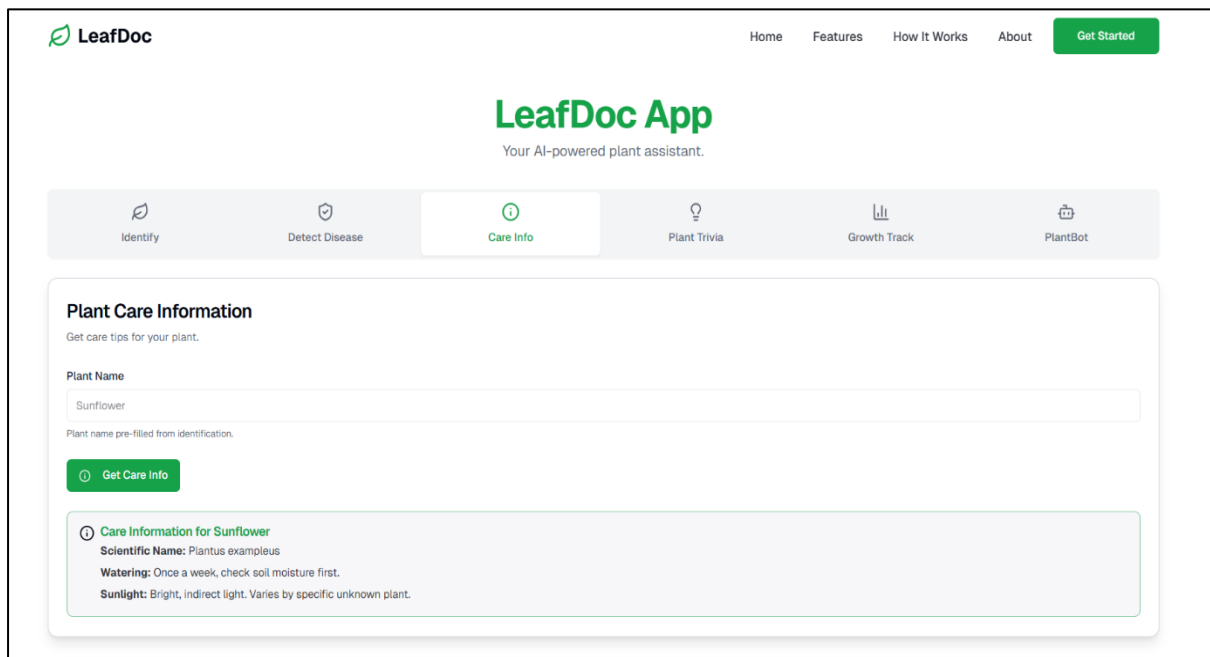


Figure 5.8: Care Information Output

6. Plant Trivia Module:

The plant trivia module provides interesting and educational facts about plants. It enhances user engagement and promotes learning about plant biology.

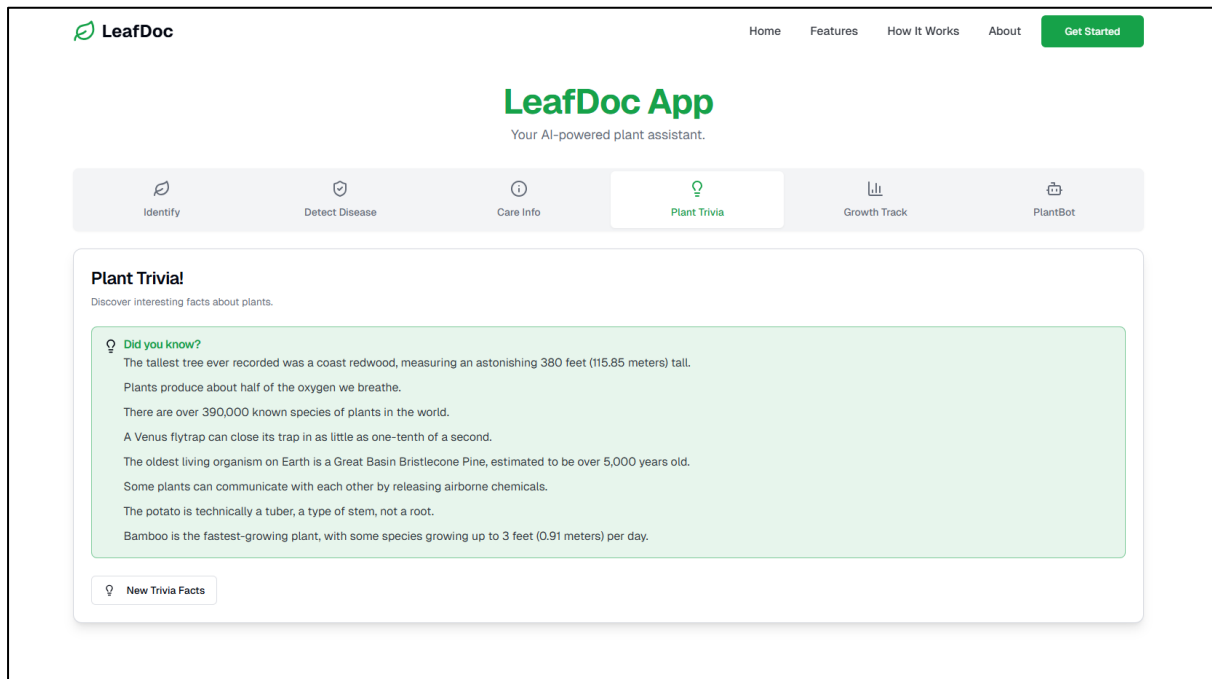


Figure 5.9: Plant Trivia

7. Growth Tracking Module:

This module allows users to monitor plant growth progress using a visual progress indicator. It helps users track development stages and maintain plant health over time.

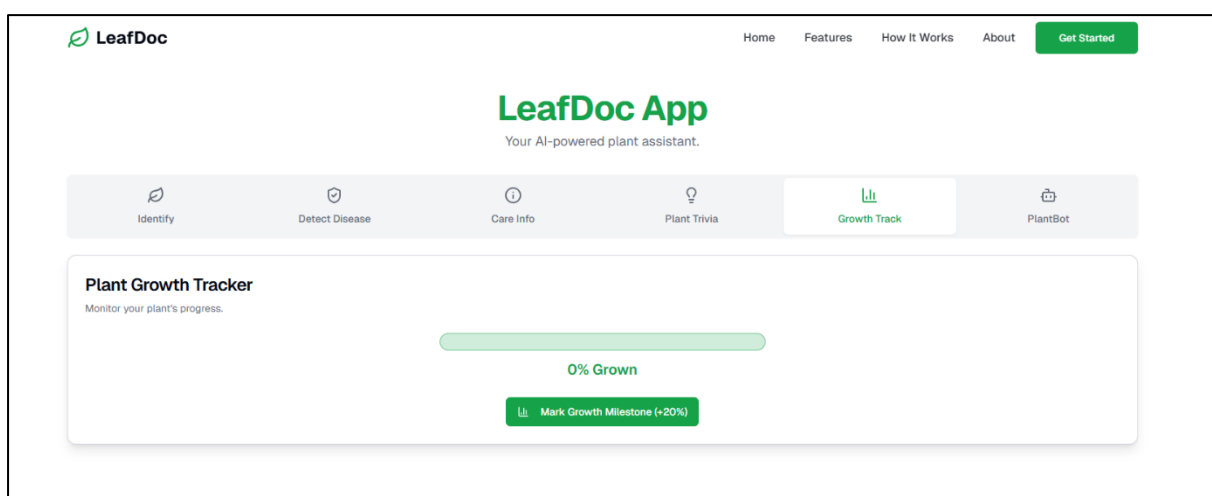


Figure 5.10: Growth Tracking Initial

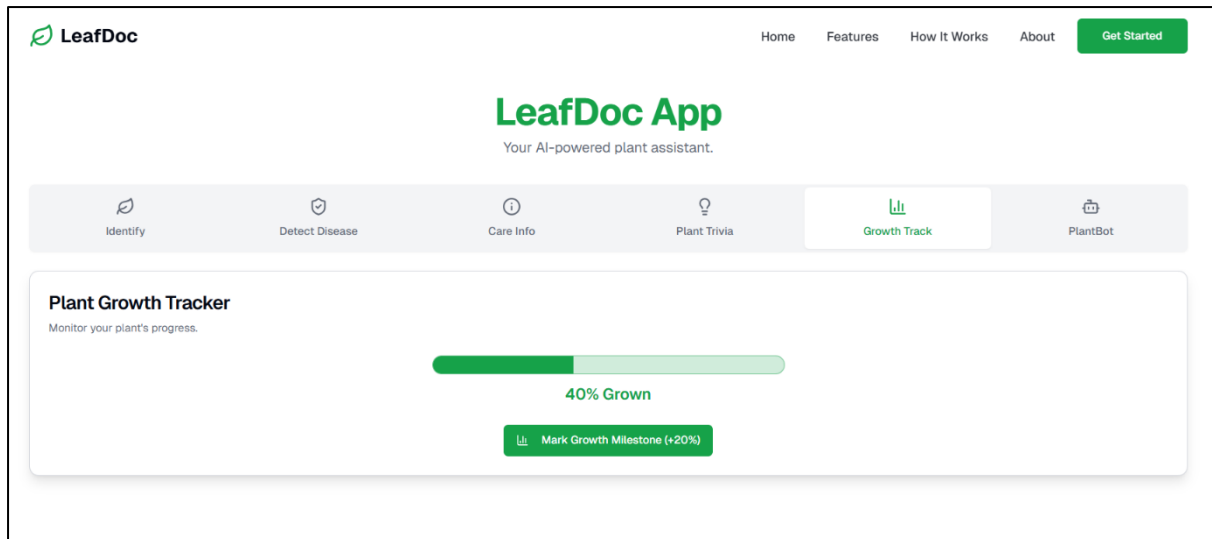


Figure 5.11: Growth Tracking

8. PlantBot (Chatbot Module):

The chatbot module provides interactive assistance by answering plant-related queries. It offers real-time guidance on plant care, diseases, and general information using AI.

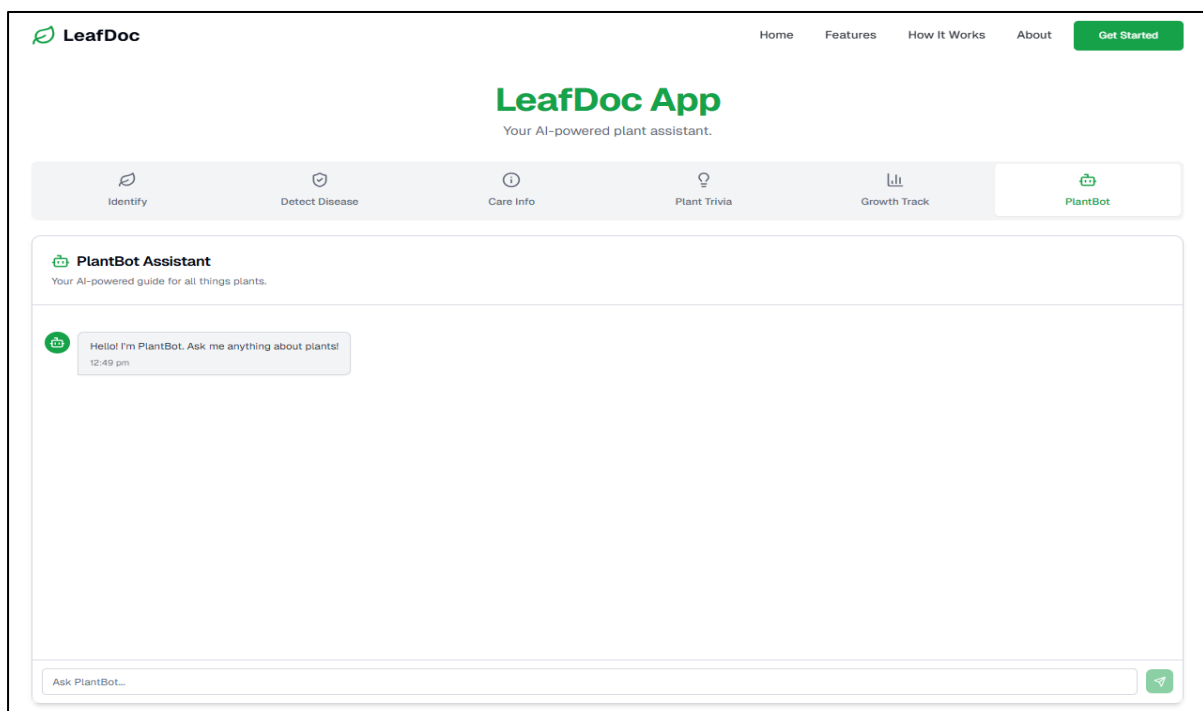


Figure 5.12: PlantBot Intial

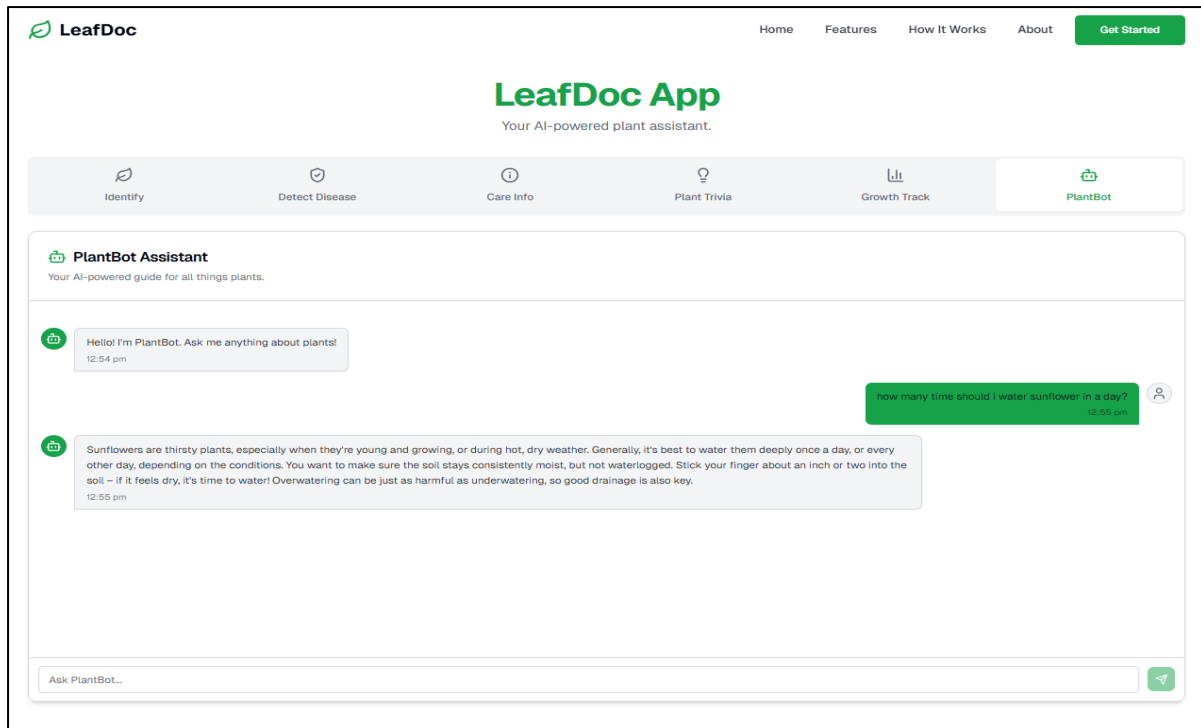


Figure 5.13: PlantBot

5.4 FUNCTIONAL RESULTS

The LeafDoc system was evaluated using multiple plant and leaf images under different conditions. For each input image, the system successfully performed plant identification and subsequently analyzed the plant for possible diseases. The results included plant species name, disease status, and relevant treatment suggestions along with care guidelines.

The disease detection module demonstrated the ability to identify visible symptoms such as discoloration, spots, and leaf damage. Based on this analysis, the system generated meaningful recommendations for treatment and plant maintenance. In addition, the chatbot module effectively handled user queries related to plant care, watering schedules, and disease prevention, providing contextual and informative responses.

Overall, the system delivered consistent and reliable outputs, ensuring a smooth and user-friendly experience across all modules.

5.5 RESULT ANALYSIS

The results indicate that the proposed system performs effectively in real-world scenarios by integrating plant identification, disease diagnosis, and advisory support into a single platform.

The use of a generative vision-language AI model enables the system to interpret plant images without requiring prior knowledge of plant species or extensive training datasets.

It was observed that the system provides accurate and meaningful outputs when the input image is clear, well-focused, and properly illuminated. The sequential workflow—starting from plant identification followed by disease detection—improves the contextual understanding of the analysis and enhances the relevance of the generated recommendations.

Unlike traditional approaches that rely heavily on predefined datasets and classification models, the proposed system demonstrates flexibility and adaptability in handling diverse plant types. This makes it particularly useful for non-expert users who may not have prior knowledge of plant species or diseases.

5.6 DISCUSSION

The implementation of LeafDoc highlights the effectiveness of integrating generative AI into plant health monitoring systems. By combining plant identification, disease detection, and intelligent advisory support, the system provides a comprehensive and user-friendly solution for plant analysis.

One of the key strengths of the system is its ability to operate without requiring manual feature extraction or large labeled datasets. This reduces development complexity and allows the system to scale easily. Additionally, the inclusion of an AI-powered chatbot enhances user interaction and provides real-time assistance, improving overall usability.

However, certain limitations were identified during testing. The accuracy of the system may vary depending on image quality, lighting conditions, and background noise. Poor-quality images can lead to less reliable results. Furthermore, the system depends on internet connectivity for API-based AI processing, which may affect performance in low-connectivity environments.

Despite these limitations, the system demonstrates strong potential as an intelligent decision-support tool for agriculture and home gardening. Its modular design allows for future enhancements, making it a scalable and practical solution for smart plant health management.

CHAPTER - VI
CONCLUSIONS AND
FURTHER SCOPE

CHAPTER 6

CONCLUSIONS AND FURTHER SCOPE

6.1 CONCLUSION

This project presented LeafDoc, a Generative AI-based plant identification and disease diagnosis system designed to assist users in monitoring plant health through image-based analysis. The system was developed to address the limitations of traditional plant diagnosis methods, which often require expert knowledge, manual observation, and multiple tools for obtaining complete information.

The proposed system successfully integrates plant identification, disease detection, and advisory support into a single web-based platform. By utilizing a generative vision-language AI model, the system is capable of analyzing plant images and generating meaningful outputs such as plant species name, disease status, and treatment recommendations. This eliminates the need for prior plant knowledge and complex model training, making the system accessible to non-expert users. The implementation results demonstrate that the system performs effectively in identifying plants and detecting diseases under various conditions. The inclusion of an AI-powered chatbot further enhances user experience by providing interactive guidance and answering plant-related queries. The modular design of the system ensures flexibility and allows future enhancements without major structural changes.

The project also highlights the practical application of artificial intelligence in agriculture and plant care. By providing a simple, efficient, and user-friendly solution, LeafDoc contributes to the development of smart agriculture systems and supports better decision-making for farmers, students, and gardening enthusiasts. Although certain limitations exist, such as dependency on image quality and internet connectivity, the system demonstrates strong potential for real-world deployment. With further improvements, it can be expanded into a more comprehensive agricultural support system.

In conclusion, LeafDoc serves as an intelligent and scalable solution for plant health monitoring, combining modern AI technologies with practical usability to address real-world challenges in agriculture and gardening.

6.2 FUTURE SCOPE

The proposed LeafDoc system demonstrates strong potential in the field of intelligent plant health monitoring. However, there are several areas where the system can be further enhanced to improve its functionality, scalability, and real-world applicability. The future scope of the system is discussed as follows:

1. Mobile Application Development

One of the major enhancements for the LeafDoc system is the development of a dedicated mobile application. A mobile platform would allow users to capture plant images directly using smartphone cameras, enabling real-time analysis in agricultural fields and home environments. This would significantly improve accessibility and make the system more practical for farmers and gardeners.

2. Real-Time Image Processing

The system can be extended to support real-time image capture and analysis. Instead of uploading static images, users could analyze plants instantly through live camera input. This feature would enhance early disease detection and allow continuous monitoring of plant health.

3. Multi-Language Support

To make the system accessible to a wider audience, especially in rural and regional areas, multi-language support can be integrated. Providing outputs such as plant identification, disease details, and chatbot responses in regional languages would improve usability and adoption among non-English-speaking users.

4. Integration with Environmental Data

Future improvements may include the integration of environmental parameters such as temperature, humidity, and soil moisture. Combining image-based analysis with environmental data can lead to more accurate and context-aware disease diagnosis and plant care recommendations.

5. Hybrid AI Approach

The system can be enhanced by combining generative AI with traditional machine learning or deep learning models trained on domain-specific datasets. This hybrid approach could improve accuracy and reliability, especially for specific crops and diseases.

6. Advanced Agricultural Advisory Features

LeafDoc can be extended to include advanced features such as fertilizer recommendations, pest detection, crop growth monitoring, and seasonal advisory services. These additions would transform the system into a comprehensive decision-support platform for smart agriculture.

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APPENDIX

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LeafDoc: A Generative AI-Based Plant Identification and Disease Diagnosis System

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Abstract: *Accurate plant identification and early disease diagnosis are essential for effective plant health management. Many existing disease detection systems assume prior knowledge of plant species, limiting their usability for non-expert users. This paper proposes LeafDoc, a Generative AI-based system that identifies a plant from an uploaded image and subsequently analyzes it for possible diseases. The system utilizes a vision-language generative AI model to generate plant identification results, disease information, and treatment recommendations. A web-based interface and integrated chatbot enhance user interaction and accessibility. Functional evaluation demonstrates that the proposed system effectively supports intelligent plant health monitoring, providing a practical and user-friendly solution for agriculture and home gardening applications.*

Keywords: Plant Identification, Plant Disease Detection, Generative AI, Vision-Language Models, Smart Agriculture

I. INTRODUCTION

Plant health monitoring is essential in agriculture and home gardening, as plant diseases can significantly reduce crop yield and quality. Early identification of plant species and timely disease diagnosis are crucial for effective plant management. However, many non-expert users lack the knowledge required to accurately identify plants, particularly during early growth stages or when only leaf images are available. Incorrect identification may result in improper treatment and economic loss. Traditional disease detection methods rely on manual inspection and expert consultation, which are time-consuming and often inaccessible in remote areas. Although recent artificial intelligence and computer vision techniques enable image-based disease detection, most existing systems assume prior knowledge of the plant species, limiting their usability. To address these challenges, this paper proposes LeafDoc, a Generative AI-based plant identification and disease diagnosis system. The system first identifies the plant species from an uploaded image and then analyzes possible diseases, generating treatment and care recommendations. Implemented as a web-based prototype using secure API-based integration of a generative vision-language model, LeafDoc provides an integrated and user-friendly solution for intelligent plant health monitoring.

II. LITERATURE REVIEW

A. Image-Based Plant Identification

Image-based plant identification has been widely studied using visual features such as leaf shape, color, and texture. Machine learning and deep learning models have demonstrated promising accuracy in recognizing plant species from leaf images. However, these approaches often require large labelled datasets or handcrafted features. Moreover, most plant identification systems operate independently and do not integrate disease diagnosis or plant health assessment.



B. Plant Disease Detection Using Image Analysis

Deep learning techniques have significantly improved plant disease detection from leaf images. Studies by Mohanty et al. and Ferentinos demonstrated the effectiveness of convolutional neural networks for disease classification. Despite high accuracy, these systems generally assume prior knowledge of plant species and focus primarily on classification performance. They rarely provide treatment recommendations or user guidance, limiting their practical usability.

C. Intelligent Advisory Systems in Agriculture

Several decision-support systems have been proposed to assist farmers in crop management and disease control. These systems often rely on rule-based logic or sensor data to generate recommendations. However, they typically lack image-based analysis and require users to manually identify plant conditions, reducing their effectiveness for non-expert users.

D. Generative AI and Vision-Language Models

Recent advancements in generative AI, particularly vision-language models, enable intelligent interpretation of images without task-specific training. These models can generate structured textual outputs from visual inputs, making them suitable for integrated applications that combine identification, diagnosis, and advisory support.

E. Research Gap

Existing research largely treats plant identification and disease detection as separate tasks, with limited integration of advisory support. The proposed LeafDoc system addresses this gap by combining plant identification, disease analysis, and AI-based recommendations into a unified, user-friendly framework.

III. METHODOLOGY

The proposed LeafDoc system is a Generative AI-based plant identification and disease diagnosis framework designed to provide integrated plant health monitoring through image analysis. The system is implemented as a web-based application using secure API integration of a vision-language generative AI model.

A. System Overview

The system operates in a sequential manner. A user uploads a plant or leaf image through the web interface. The image is analyzed by a generative AI model, which first identifies the plant species and subsequently evaluates the image for potential diseases. Based on the analysis, the system generates treatment recommendations and plant care guidance. An AI-powered chatbot enhances user interaction and provides contextual assistance.

B. System Architecture

The architecture comprises the following modules:

- User Interface Module
- AI Image Analysis Module
- Plant Identification Module
- Disease Diagnosis Module
- Recommendation Module
- Chatbot Module

These components work collaboratively to ensure accurate identification, diagnosis, and advisory support.

C. Operational Workflow

The methodology follows five structured stages:

- Image acquisition through web interface

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- AI-based plant identification
- Disease detection based on visual symptoms
- Generation of treatment and care recommendations
- Interactive chatbot support

This integrated approach eliminates the need for manual feature extraction or plant-specific model training and provides a unified solution for intelligent plant health monitoring.

IV. IMPLEMENTATION AND RESULTS

A. System Implementation

The proposed LeafDoc system is implemented as a web-based prototype following a client-server architecture. The frontend is responsible for user interaction, while the backend manages image processing and AI-based analysis through secure API integration.

The user interface allows users to upload plant or leaf images and view structured analysis results. Upon image submission, the backend securely communicates with a generative vision-language AI model to perform plant identification and disease diagnosis. Separate processing workflows are maintained for identification, diagnosis, and chatbot interaction to ensure modularity and scalability.

Sensitive configuration parameters, including API credentials, are managed using environment variables to maintain system security. The prototype implementation demonstrates the feasibility of integrating generative AI into a practical plant health monitoring application.

B. Functional Results

The system was evaluated using multiple plant and leaf images to verify functional performance.

1. Plant Identification

When a plant image was uploaded, the system successfully identified the plant species and generated structured output including the species name and confidence level.

As shown in Fig. 1, the system correctly identified a sunflower image as *Helianthus annuus* with a confidence score of 95%. This demonstrates the capability of the AI model to interpret visual features and provide reliable plant identification results.

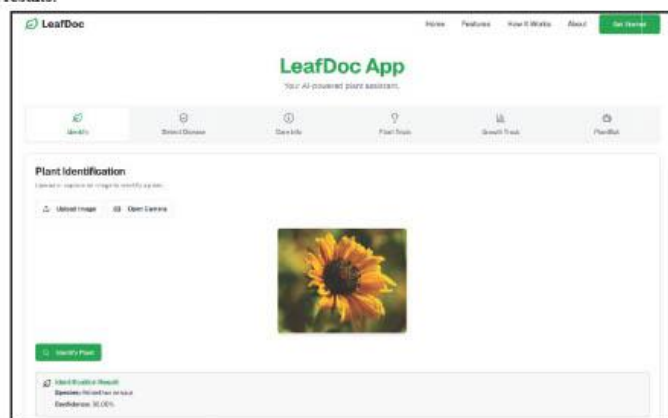


Fig. 1. Plant Identification Result

2. Disease Detection and Recommendation

Following plant identification, the system performed disease analysis on symptomatic leaf images.

As illustrated in Fig. 2, the system detected Rose Rust disease with a confidence score of 85% based on visible symptoms such as yellowing and brown spots. In addition to identifying the disease, the system generated practical treatment suggestions, including removal of infected leaves and use of appropriate fungicides.

This integrated output highlights the system's ability to combine identification, diagnosis, and advisory support within a single workflow.



Fig. 2. Disease Detection and Recommendation Output

C. Result Analysis

The experimental results indicate that the proposed system successfully performs sequential plant identification and disease diagnosis using a single generative AI framework. The evaluation focuses on functional correctness and usability rather than traditional dataset-based accuracy metrics, as the system relies on API-based generative inference.

The results demonstrate:

- Accurate plant species identification
- Effective detection of visible plant diseases
- Generation of meaningful treatment recommendations
- Seamless integration between analysis and user interface

The system proved particularly useful for non-expert users by eliminating the requirement for prior plant knowledge before disease diagnosis.

D. Discussion

The implementation confirms that integrating plant identification and disease detection within a unified generative AI framework enhances system usability and practicality. The API-based approach allows the system to analyze a wide variety of plants without requiring model retraining or large labeled datasets.

However, system performance may depend on factors such as image clarity, lighting conditions, and symptom visibility. Despite these limitations, the results demonstrate strong potential for real-world application in agriculture and home gardening. The modular architecture also enables future enhancements, including multilingual support, environmental data integration, and large-scale deployment.

V. CONCLUSION

This paper presented LeafDoc, a Generative AI-based plant identification and disease diagnosis system for intelligent plant health monitoring. The system first identifies plant species from uploaded images and then performs disease analysis, providing treatment and care recommendations through an integrated web-based platform. Functional evaluation demonstrated that the system successfully performs plant identification and disease detection while offering user-friendly advisory support through an AI-powered chatbot. The proposed approach eliminates the need for prior plant knowledge and complex model training, making it suitable for non-expert users. Future enhancements may include mobile application integration, multilingual support, and environmental data incorporation to further improve system scalability and practical usability.

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CHAPTER 1 INTRODUCTION 1.1 OVERVIEW The LeafDoc is a web-based application designed to assist users in plant identification and disease diagnosis using artificial intelligence. In agriculture, horticulture, and home gardening, maintaining plant health is essential for ensuring productivity, sustainability, and environmental balance. Plants are highly susceptible to various diseases caused by bacteria, fungi, viruses, and environmental factors, which can significantly reduce crop yield and quality if not detected at an early stage. Traditionally, plant identification and disease diagnosis rely on manual observation and expert knowledge. Farmers and gardeners often depend on agricultural experts or personal experience to identify plant species and diagnose diseases. However, this process can be time-consuming, costly, and not always accessible, especially in rural or remote areas where expert consultation may not be readily available. Moreover, incorrect identification or delayed diagnosis can lead to improper treatment, resulting in plant damage and economic loss. With the advancement of digital technologies, there has been a growing interest in developing automated systems for plant analysis. Image-based analysis using artificial intelligence has emerged as a powerful approach for recognizing plant species and detecting diseases. These systems utilize visual features such as leaf shape, color, texture, and patterns to identify plants and analyze their health condition. However, many existing solutions require prior knowledge of the plant species or depend on large, pre-trained datasets, which limits their usability for general users. The LeafDoc system addresses these limitations by utilizing a generative vision-language AI model capable of interpreting plant images and generating meaningful outputs. The system allows users to upload images of plant leaves through a simple web interface. Based on the uploaded image, the system first identifies the plant species and then analyzes the image to detect possible diseases. It provides detailed information, including plant name, disease status, treatment suggestions, and plant care guidelines. One of the key features of LeafDoc is the integration of an AI-powered chatbot that enables interactive communication between the user and the system. Users can ask questions related to plant care, disease prevention, and treatment methods, and the chatbot provides contextual and informative responses. This feature enhances user engagement and makes the system more accessible, especially for non-expert users. The system is designed to be user-friendly, scalable, and efficient. It eliminates the need for manual feature extraction or complex model training by leveraging API-based generative AI. This approach simplifies the implementation while maintaining high flexibility and adaptability. The web-based nature of the system ensures that users can access it from different devices without requiring installation. Overall,

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Course Outcome:

CO1: Implement comprehensive project planning, designing, and development processes.

CO2: Acquire and understand software industry needs and requirements.

CO3: Analyze real-world problems and develop appropriate software solutions using modern tools and technologies.

CO4: Demonstrate teamwork, effective communication, and professional and ethical responsibilities during project development.

Program Outcome

	Program Outcome
PO1	Computational Knowledge: Apply knowledge of computing fundamentals, computing specialisation, mathematics, and domain knowledge appropriate for the computing specialisation to the abstraction and conceptualization of computing models from defined problems and requirements.
PO2	Problem Analysis: Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
PO3	Design /Development of Solutions: Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PO4	Conduct Investigations of Complex Computing Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5	Modern Tool Usage: Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.
PO6	Professional Ethics: Understand and commit to professional ethics and cyber regulations, responsibilities, and norms of professional computing practice.
PO7	Life-long Learning: Recognise the need, and have the ability, to engage in independent learning for continual development as a computing professional.
PO8	Project Management and Finance: Demonstrate knowledge and understanding of the computing and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PSO 1: Solve real-world computing problems of various industries by understanding and applying computing techniques with other related disciplines

PSO 2: Design, develop, test and maintain desktop, website, mobile and cross-platform software applications using modern tools, technologies & skills

CO_PO Mapping										
Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PS01	PS02
CO1	3	3	2	3	3	3	3	3	3	3
CO2	2	3	3	3	2	3	3	3	3	2
CO3	3	3	3	2	3	2	3	3	3	3
CO4	3	2	3	3	3	3	3	3	2	2